WEEK 1 PROJECT :

MACHINE LEARNING

Dataset Name : Mobile Price Prediction cleaned_data

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
print("---- Load Dataset ----")
data = pd.read_csv("Mobile Price Prediction cleaned_data.csv")
print("\nDataset Preview:")
print(data.head())
print("\nDataset Info:")
print(data.info())
print("\n---- Preprocessing ----")
X = data[['RAM']]  # Feature
y = data['Actual Price Range'] # Target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print("\n---- Sklearn Linear Regression ----")
model = LinearRegression()
model.fit(X_train, y_train)
y_pred_sklearn = model.predict(X_test)
mse_sklearn = mean_squared_error(y_test, y_pred_sklearn)
print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
print("MSE:", mse_sklearn)
```

```
print("\n---- Manual Linear Regression ----")
x = X_train.values.flatten()
y_manual = y_train.values
n = len(x)
sum_x = np.sum(x)
sum_y = np.sum(y_manual)
sum_xy = np.sum(x * y_manual)
sum_x2 = np.sum(x ** 2)
m = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
c = (sum_y - m * sum_x) / n
x_test = X_test.values.flatten()
y_pred_manual = m * x_test + c
mse_manual = mean_squared_error(y_test, y_pred_manual)
print("Slope (m):", m)
print("Intercept (c):", c)
print("MSE:", mse_manual)
print("\n---- Comparison Table ----")
comparison = pd.DataFrame({
    "RAM": x_test,
    "Actual Price Range": y_test.values,
    "Sklearn Prediction": y_pred_sklearn,
    "Manual Prediction": y_pred_manual
})
print(comparison.head())
comparison.to_csv("Mobile Price Prediction cleaned_data.csv", index=False)
print("\nPredictions saved to Mobile Price Prediction cleaned_data.csv")
```

OUTPUT:

```
PS C:\Users\Lenovo\Desktop\ML week1> & C:\Users\Lenovo\AppData\Local\Programs\Python\Python313\python.exe
o/Desktop/ML week1/prg.py"
---- Load Dataset ----

Dataset Preview:
   RAM  Actual Price Range  Sklearn Prediction  Manual Prediction
0  6.0                 982         9390.289157        9390.289157
1  6.0               16999         9390.289157        9390.289157
2  4.0                9999         5402.060241        5402.060241
3  3.0                9990         3407.945783        3407.945783
4  8.0               31590        13378.518072       13378.518072

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 4 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   RAM                 7 non-null      float64
 1   Actual Price Range  7 non-null      int64
 2   Sklearn Prediction  7 non-null      float64
 3   Manual Prediction   7 non-null      float64
dtypes: float64(3), int64(1)
memory usage: 356.0 bytes
None
```

```
---- Preprocessing ----

---- Sklearn Linear Regression ----
Slope (m): 39.86538461538476
Intercept (c): 16772.742307692304
MSE: 128479485.5388905

---- Manual Linear Regression ----
Slope (m): 39.86538461538461
Intercept (c): 16772.742307692308
MSE: 128479485.53889056

---- Comparison Table ----
   RAM  Actual Price Range  Sklearn Prediction  Manual Prediction
0  6.0                 982        17011.934615       17011.934615
1  6.0               16999        17011.934615       17011.934615

Predictions saved to Mobile Price Prediction cleaned_data.csv
PS C:\Users\Lenovo\Desktop\ML week1>
```